INTERNATIONAL SEARCH REPORT

International application No.
PCT/JP2022/038211

A. CLASSIFICATION OF SUBJECT MATTER

G01N 3/56(2006.01)i
FI: G01N3/56 H

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

G01N3/00-3/62; C01B32/25

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Published examined utility model applications of Japan 1922-1996
Published unexamined utility model applications of Japan 1971-2022
Registered utility model specifications of Japan 1996-2022
Published registered utility model applications of Japan 1994-2022

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

JSTPlus/JST7580 (JDreamIII)

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	李真和,真鍋佳典,寺本三記,原野佳津子,小林豊,角谷均. “各種ダイヤモンドの摩耗特性”. ２０１９年度精密工学会春季大会学術講演会講演論文集. 13 March 2019, pp. 646, 647, Internet: <URL: https://doi.org/10.11522/pscjspe.2019S.0_646>, (LEE, Masakazu, MANABE, Keisuke, TERAMOTO, Minori, HARANO, Katsuko, KOBAYASHI, Yutaka, SUMIYA, Hitoshi. Wear characteristics of various diamonds), non-official translation (Proceedings of the 2019 JSPE Spring Meeting.) entire text, all drawings	1-7
A	WO 2019/069888 A1 (KUSANO, Eiji) 11 April 2019 (2019-04-11) entire text, all drawings	1-7
A	JP 2014-91661 A (SUMITOMO ELECTRIC IND. LTD.) 19 May 2014 (2014-05-19) entire text, all drawings	1–7
A	US 2011/0146372 A1 (VAREL EUROPE S.A.S.) 23 June 2011 (2011-06-23) entire text, all drawings	1–7
A	CN 105158098 A (NANJING UNIVERSITY OF AERONAUTICS AND ASTRONAUTICS) 16 December 2015 (2015-12-16) entire text, all drawings	1–7

☐ Further documents are listed in the continuation of Box C. ☑ See patent family annex.

* Special categories of cited documents:
“A” document defining the general state of the art which is not considered to be of particular relevance
“E” earlier application or patent but published on or after the international filing date
“L” document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)
“O” document referring to an oral disclosure, use, exhibition or other means
“P” document published prior to the international filing date but later than the priority date claimed
“T” later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention
“X” document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone
“Y” document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art
“&” document member of the same patent family

Date of the actual completion of the international search	Date of mailing of the international search report
15 November 2022	**29 November 2022**
Name and mailing address of the ISA/JP **Japan Patent Office (ISA/JP)** **3-4-3 Kasumigaseki, Chiyoda-ku, Tokyo 100-8915** **Japan**	Authorized officer Telephone No.

INTERNATIONAL SEARCH REPORT
Information on patent family members

International application No.
PCT/JP2022/038211

Patent document cited in search report	Publication date (day/month/year)	Patent family member(s)	Publication date (day/month/year)
WO 2019/069888 A1	11 April 2019	JP 2019-66366 A	
JP 2014-91661 A	19 May 2014	(Family: none)	
US 2011/0146372 A1	23 June 2011	US 2013/0239652 A1	
		US 2011/0146373 A1	
		US 2011/0146374 A1	
		US 2011/0148021 A1	
		WO 2011/075432 A2	
		WO 2011/075435 A1	
		WO 2011/075438 A1	
		WO 2011/075479 A1	
CN 105158098 A	16 December 2015	(Family: none)	